Here is the detailed summary of the meeting, categorized by paper pattern, unit-wise important topics, and question types:

1. Examination Pattern & Overall Breakdown

- **Ratio of Questions**: Approximately **70:30** or **75:25** between Theoretical/Application-based questions and Mathematical/Calculation-based questions.
- **Weightage of Numericals**: Numerical questions will comprise **20% to 25%** of the paper.
- **Target Marks**: The professor mentioned it is easily possible to score 99–100 marks if answers stick to the point and meet the merit of the question.
- **Answer Writing Tip**: Focus on answering according to the **merit of the question**; avoid filling unnecessary pages (e.g., writing 5 to 10 pages per answer is not required).

2. Mathematical & Numerical Questions

You can expect **2 to 3 numerical problems** in the short/two-mark questions and a few in the long-answer section:

Net Productivity:

$$NetProductivity = GrossPrimaryProductivity - Respiration$$

- **Sound Power & Pressure Level**: Calculations based on the 2 standard formulas for Sound Power Level (L_W) and Sound Pressure Level (L_P).
- **Biological Oxygen Demand (BOD)**: Problems involving the 2–3 modified forms of the BOD formulas.
- **Vehicle & Population Emissions**: Calculations predicting emission levels per person/vehicle over time (e.g., daily to yearly conversion).
- **Hydrocarbon Emissions/Discharge**: Problems given in the extra question set provided in the document from HOD.
- **Energy Flow / Trophic Levels**: Calculations based on the **10% Energy Law** across trophic levels (e.g., calculating energy retained or released to the atmosphere from T_1 to T_4).

3. Detailed Unit-Wise Breakdown

Unit 1 & Unit 2: Cell Structure, Biomolecules, & Biochemistry

- **Cell Structure**:
  - Characteristics of living beings (Potential long question).
  - Detailed structural comparison between **Animal Cell, Plant Cell, and Bacterial Cell**.
- **Proteins**:
  - Structural classification of proteins: Primary, Secondary, Tertiary, and Quaternary structures along with their significance.
- **Carbohydrates**:
  - Classification of carbohydrates.
- **Enzymes & Pesticides**:
  - Types and classification of enzymes.

- Classification, advantages, and disadvantages of pesticides (7–8 types of pesticides).
- Biological pest control methods.

- **Vaccines & Bio-energy**:
- Types of vaccines.
- Bio-energy: Types, advantages, and disadvantages.

Unit 3: Ecosystems & Environmental Dynamics

- **Food Chains & Food Webs**: Definitions and short questions.
- **Trophic Levels & Energy Flow**: Energy flow through ecosystems and how it relates to the **1st and 2nd Laws of Thermodynamics**.
- **Biogeochemical Cycles**: Focus on drawing and explaining all steps for:

1. **Nitrogen Cycle** *(Most important & tricky, 4–5 steps)*
2. **Carbon Cycle**
3. **Oxygen Cycle**
4. **Hydrological (Water) Cycle**

- **Environmental Gradient Curve**: Explaining population levels across the curve (under-population, optimal population, and over-population).
- **Environmental Impact Assessment (EIA)**: The 4–5 key steps involved in EIA (Scoping, Planning, Assessment, etc.).

Unit 4: Air & Noise Pollution

- **Noise Pollution**:
- Causes, types of noise levels, and health effects of noise.
- Noise control techniques and noise management.
- **Noise Standards (2000)** (Expected as a 2-mark short question).

- **Air Pollution & Global Issues**:
- Acid Deposition Mechanism.
- **Greenhouse Effect & Global Warming**: How greenhouse gases impact global climate change (Long question format).
- Hydrocarbon discharge and vehicle emission calculations.

Unit 5: Water Pollution & Waste Management

- **Solid & Hazardous Waste Management**:
- **Integrated Solid Waste Management (ISWM)**.
- **E-Waste Management**.
- Definition, classification, and management of Hazardous Waste.
- The different **R's** involved in waste management systems.
- **Carbon Sequestration / CCS**: Different processes and techniques.

- **Water Treatment Processes**:
- Comparison between **Primary Water Treatment** and **Tertiary Water Treatment**.
- Mechanisms of **Coagulation, Flocculation, and Sedimentation**.
- Disinfection and Sterilization techniques.

4. Short / Two-Mark Question Topics

- Definitions of **Peptide bond**, **Glycosidic linkage**, **Dipeptide**, **Tripeptide**, and **Oligosaccharides**.

- Transmembrane proteins: Definition, significance, and functions.
- Vitamin and mineral deficiency diseases (e.g., Vitamin A, Vitamin B9).
- Metal centers in biological structures:
  - **Chlorophyll**: Magnesium (Mg).
  - **Vitamin B12**: Cobalt (Co).
  - **Vitamin D3 / Carboxylase**: Calcium (Ca^{2+}, oxidation state $+2$).
- Advantages of **Bio-fertilizers** and **Bio-fuels**.
- Definition and noise limit values under **Noise Standards 2000**.